

AutoSampler

Production & Publication Readiness Review

An independent code review of the **main** branch, conducted from the dual perspective of a code reviewer and a prospective user, ahead of a research publication announcing the package.

Review date: 2026-06-30 | Method: five parallel domain reviewers (code correctness, documentation, tests/CI/packaging, examples/UX, test-coverage map) plus direct lead verification of all headline findings.

Verdict

Strong scientific core; not yet release-ready as found. The algorithmic layer (MSM estimation & convergence, adaptive binning, weighted ensemble, VAMP-2 feature selection, retraining, scheduler logic) is well-engineered and genuinely well-tested, and the docs site, annotated template, and rendered notebook are excellent for a research package. However, the review found **~6 release-gating blockers** and a band of correctness, reproducibility, packaging, and example/test gaps. A subset of blockers has already been fixed during this review (Section 2); the remainder is itemised with a prioritised plan.

Area	As-found state	Severity band
Release gates	Test suite RED; documented install broken; Quick Start example could not run	Blocker
Correctness / robustness	Walker-failure & hang handling, engine/scheduler edge cases, target-mode crash	Major
Reproducibility	Seeding not actually deterministic across the training loop (matters for a paper)	Major
Packaging / release	Incomplete PyPI metadata, heavy deps, partial CI lint, no release/DOI	Major
Docs / examples / tests	Gaps: API ref, citations, several features undemoed, orchestration untested	Major / Minor

Scope note: findings are rated Blocker / Major / Minor / Nit. CONFIRMED items were traced in source by the lead reviewer; others are reviewer-reported with a code citation. One widely-suspected bug was investigated and dismissed as a false positive (see Section 6).

1. What is already strong

- **MSM subsystem** (estimator + convergence monitor): connected-set restriction, MLE/Bayesian estimation, implied timescales, VAMP-2, PCCA+, and a flux-weighted transition-matrix convergence criterion — with real, behaviour-asserting tests (recovers 3-state systems, serialization round-trips).
- **Adaptive binning** (gradient / mab / eigenvector), **weighted ensemble** (weight-conserving split/merge), **VAMP-2 feature selection**, and **retraining policies** are cleanly factored and tested.
- **Execution backends** (local / SLURM / PBS) behind a clean strategy interface; scheduler logic is unit-tested with a fake command runner.
- **Developer experience**: annotated input-file template + autosampler-init, a full MkDocs site, and a rendered Jupyter notebook tutorial that runs on synthetic data without a GPU.
- Test quality where present is high — assertions check real numerical/behavioural outcomes, not merely the absence of exceptions.

2. Fixed during this review

The following were implemented and pushed to branch `claude/autosampler-analysis-plan-jj0n1q` (suite green at 101 tests). They are recorded here so the report doubles as a change log.

#	Item	Resolution
1	Test suite RED — examples/template.yaml drifted from the module template	Regenerated from the single source; suite green.
2	Documented conda install broken (env.yml pinned pydantic 1.x; missing shapely)	Bumped to pydantic>=2.0, added shapely + deep-CV backends.
3	Delta-checkpoint resume truncated history (broke autosampler-path); non-atomic writes	Reconstruct full history across deltas; atomic writes; tolerate a corrupt delta; added regression tests.
4	Local backend aborted the whole iteration if one walker failed	Catch per-walker failure, log, mark unsuccessful (matches scheduler path).
5	Incomplete PyPI metadata; version duplicated	Added readme/license/urls/keywords/authors; single-sourced version; Beta status.
6	No citation infrastructure	Added CITATION.cff + README 'How to cite' (author list / ORCIDs / DOI flagged as TODO).
7	AlaD Quick Start unrunnable (hardcoded dev path; silent platform typo)	Removed the path, fixed platform_name; documented the GROMACS FF requirement.
8	Doc drift (blob/devel link; phi_psi advertised as generic)	Fixed link to main; documented phi_psi as AIB9-specific.

3. Release-gating blockers

Must be resolved before a general release. Items 1–4, 6 below are addressed in Section 2; the rest remain open.

Blocker	Detail & status
Red test suite	tests/test_input_template.py failed on main. FIXED .
Broken documented install	env.yml pydantic 1.x vs Pydantic-v2 code; missing shapely. FIXED .
Delta-checkpoint regression	autosampler-path read a truncated (delta-only) history; non-atomic writes. FIXED .
Quick Start cannot run	AlaD needs an external GROMACS force field; no CPU-only hello-world exists. PARTIAL — de-risked; a self-contained alanine-dipeptide example is the agreed next step.
phi_psi crashes off-AIB9	adaptive_feature_type: phi_psi hard-requires 9 AIB residues but is advertised generically. Documented ; code still AIB9-only — rename or generalise.
No citation / DOI	Gates citability for the paper. Scaffolded (CITATION.cff); needs the confirmed author list, ORCIDs, affiliations, and a Zenodo DOI.

4. Major — correctness & robustness

Finding	Risk	Location
No MD timeout / watchdog on the default (OpenMM, in-process) engine or the local backend; scheduler poll loop has no overall deadline	A hung GPU/driver stalls the campaign forever	execution/local.py; engines/openmm.py; scheduler.py
OpenMM engine only returns True or raises; NaN-recovery re-steps outside any guard	An unstable spawn crashes the worker instead of reporting failure	engines/openmm.py
Subprocess engines move output after exit-0 without checking a trajectory was produced	mdrun exiting 0 with no/empty output → uncaught error	engines/gromacs.py; amber.py
Scheduler treats a failed poll command as 'job done'; SLURM job-id match is a \b-bounded substring (123 vs 1234)	Transient queue/qstat hiccup abandons a healthy job; id collision	execution/scheduler.py ; slurm.py
expected_frames fallback + bare except in frame mapping; only the aggregate count is checked	Offsetting over/under-production mis-assigns CV rows → corrupt lineage	paths.py
Density/Voronoi_weighted_choice uses replace=False with many zero weights (target mode)	np.random.choice raises 'fewer non-zero entries than size'	spawners/density.py
Triclinic/truncated-octahedron boxes reduced to box diagonal + 90°	Wrong PBC for common Amber solvated systems, silently	engines/amber.py; gromacs.py
deep-tica projection runs on CPU tensors while params may be on CUDA; no NaN/convergence guard in CV training	Device-mismatch RuntimeError; NaN CVs flow silently into binning/spawning	spaces/model.py; spib.py

5. Major — reproducibility, packaging, docs/examples/tests

Reproducibility (matters for the publication's claims)

- Seed is set once at startup, not before each CV retrain — 2nd+ retrains depend on all intervening RNG draws.
- `torch.use_deterministic_algorithms` is never enabled; `tvae/vampnet` DataLoaders get no seeded generator (`vampnet` shuffles on the global RNG).
- SPIB hardcodes `seed=42` and resets the global torch RNG mid-loop, ignoring `random_seed` and perturbing later draws.
- Density/Voronoi/LOF/FPS spawners and the Voronoi binner draw from the unseeded global `np.random`. Net: runs are not bit-reproducible despite the `SeedManager`.

Packaging / release (target: PyPI + conda)

- Heavy deps are mandatory; `openmm` is effectively conda-only, so a plain pip install often fails to resolve — move `openmm` (and likely `MDAnalysis`) to extras with a lazy import so the base install works.
- CI lints only ~18 hand-picked files; the most-edited modules (`core.py`, `config.py`, `engines/*`, `checkpoints/manager.py`) are unlinted (~250 ruff findings incl. from `openmm import * star imports`). Lint the whole tree.
- No coverage gate, no `mkdocs build --strict` check, no tag-triggered release/Zenodo workflow; Python 3.12 advertised but untested.
- Version single-sourcing & full metadata: **DONE** in Section 2.

Docs, examples & tests

- Config schema uses the Pydantic default (`extra=ignore`): a typo'd key passes `--check` silently — consider `extra=forbid` (and fix the example configs that rely on it).
- `configuration.md` omits several real keys; CLI reference omits `autosampler-init / -analyze`; no API reference (`mkdocstrings`); method citations are thin (TICA/VAMPNet/PCCA+/MAB uncited).
- No worked example for `spib`, `deep-tica`, `lof`, `fps`, `we`, target mode, `pbs`, or `mab/eigenvector` binning — several are headline features.
- `core.py` orchestration, MD engines, half the spawners (`voronoi/lof/fps`), `paths.py`, and 3/5 CLIs are effectively untested; no end-to-end iteration test. (Delta checkpointing is now covered.)

6. Minor / nits & a dismissed false positive

- Leftover debug `print("EXACT SPAWN INDICES", ...)` (`core.py`); hardcoded personal `gmx` path (`paths.py`); dead `hint` variable (`openmm.py`).
- FPS can return duplicate indices; TVAE `BatchNorm` + a final batch of size 1 raises; `binning.find_bins` raises `AttributeError` if called before fit.
- Broad `except Exception` swallowing without logging in several spawner/binner fallbacks; unbounded `scheduler_jobs/` artifact growth.
- No NaN/inf validation of input CV points across spawners.

Dismissed false positive. A reviewer flagged the MSM least-counts weight as $1/\text{count}^2$. On tracing it, the per-frame value correctly distributes the microstate-level $1/\text{count}$ least-counts weight across that microstate's frames — exactly mirroring the MSM-guided path's `base / size_per_frame`. Left unchanged.

7. Recommended path to release

Priority 1 — finish the blockers

- Ship the self-contained CPU-only alanine-dipeptide hello-world; generalise or rename `phi_psi`; confirm the author list + mint a Zenodo DOI in `CITATION.cff`.

Priority 2 — correctness & reproducibility

- Add an MD-walker timeout; make the OpenMM engine report failure instead of raising; verify trajectories were produced; harden scheduler polling.
- Plumb the configured seed through every fit / spawner / DataLoader and enable torch determinism — required to claim reproducibility in the paper.

Priority 3 — packaging, docs, examples, tests

- Move `openmm/MDAnalysis` to extras with lazy imports; lint the whole tree in CI; add coverage + a strict docs build + a release/Zenodo workflow; test on 3.12.
- Add an API reference + a references page; complete the config & CLI docs; add examples for the undemoed features; broaden tests to orchestration, engines, the remaining spawners, paths, and CLIs, plus one end-to-end iteration test.

Bottom line: the science and architecture are publication-grade; the gap to release is engineering hardening, reproducibility, packaging, and coverage — all tractable, and partly already underway on the review branch.